A TORSION FREE COHERENT MODULE ON AN INTEGRAL REGULAR CURVE IS FINITE LOCALLY FREE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file divisors.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Carrier window: statement and proof both lie inside the window 0-2000 granted to this lane by CONTROL 8.83.

Editorial note. The card was recovered after a quality read: the prefilter dropped the candidate for the phrase 'It is clear that a finite locally free module is torsion free', which only dismisses the easy direction of an if-and-only-if whose converse is argued in full.

Editorial note. The three references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: sources/stacks-divisors/retained-preamble.tex (88 lines, verbatim present).

Allowed standard prerequisites (editorial).

- Finitely presented modules with free stalks are locally free. Algebra, Lemma algebra-lemma-finite-projective, used at proof lines 1812-1815. Turns freeness of the stalks into finite local freeness.
- A regular local ring of dimension one is a discrete valuation ring. Algebra, Lemma algebra-lemma-characterize-dvr, used at proof lines 1818-1820. Identifies the local rings in the one-dimensional case.
- A finite torsion free module over a Dedekind domain is free. More on Algebra, Lemma more-algebra-lemma-dedekind-torsion-free-flat, used at proof lines 1820-1822. Gives freeness of the stalk over the DVR.

Source reference. Original lemma 00PD: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 927–927. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 00NX: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 875–875. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 0AUW: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 13664–13664. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let X be an integral regular scheme of dimension ≤ 1. Let $\mathcal{F}$ be a coherent $\mathcal{O}_X$-module. The following are equivalent*

Date: stacks-project:divisors.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

(1) $\mathcal{F}$ *is torsion free,*
(2) $\mathcal{F}$ *is finite locally free.*

Proof. It is clear that a finite locally free module is torsion free. For the converse, we will show that if $\mathcal{F}$ is torsion free, then $\mathcal{F}_x$ is a free $\mathcal{O}_{X,x}$-module for all $x \in X$. This is enough by Algebra, Lemma 00NX and the fact that $\mathcal{F}$ is coherent. If $\dim(\mathcal{O}_{X,x}) = 0$, then $\mathcal{O}_{X,x}$ is a field and the statement is clear. If $\dim(\mathcal{O}_{X,x}) = 1$, then $\mathcal{O}_{X,x}$ is a discrete valuation ring (Algebra, Lemma 00PD) and $\mathcal{F}_x$ is torsion free. Hence $\mathcal{F}_x$ is free by More on Algebra, Lemma 0AUW. □